

BRCMAP.NET · BLACK ROCK CITY 2026 · "AXIS MUNDI"

AFTER ACTION REPORT

One unofficial map, one event week, four ways of seeing it: a traffic engineer, a participant, the Organization, and law enforcement. What a community-run map with GPS, mesh radio, live weather, streaming radio, and a crowd-sensed exodus pulse actually did – and honestly did not do – for a temporary city of roughly seventy thousand.

Prepared September 8, 2026, at the close of Exodus; revised September 10 with every figure re-verified against production data and contemporary reporting · BRC Map is an independent community project, unaffiliated with Burning Man Project · brcmap.net

EXECUTIVE SUMMARY

BRC Map entered event week as a volunteer-run, tile-free map holding 236 camps that had registered themselves. It left the week as a live operational picture of the whole city: **1,247 camps** (236 community-placed, 1,011 geocoded from Burning Man's public placement directory), **361 placed artworks**, 10,240 event listings across five guides, live weather from sensors inside the fence, an app-wide radio player, a rideshare board with consent-scoped live-location rendezvous, a crowd-sensed Gate Road traffic grade, and – as Restoration began – an anonymous MOOP-marking tool. It served **440 registered accounts** and a measured peak of **239 distinct visitors in a day** (September 5 – a floor, not a count: the metric is deliberately blind behind shared connections). It ran the entire event with **zero downtime**, entirely offline-capable, with no third-party tracker of any kind.

The unifying design decision, visible from every perspective below, is that the map **tells the truth at the edge of its knowledge**: measurements are shown with their evidence, grades disappear when their data goes stale, and nothing is ever invented to fill a silence.

MEASURE	VALUE	NOTE
Camps on the map	1,247	236 self-placed · 1,011 from the public placement directory
Directory geocoding agreement	99% ≤ 150 m	median 37 m vs. camps' own pins, where both existed
Artworks placed	361	from 9 before the official art directory import
Registered accounts	440	
Peak measured daily visitors	239	Sep 5; anonymous – undercounts shared connections by design
Busiest pages	/live · / · /rides	weather, the map, and rides – in that order
Changes shipped, gate to Resto	39 pull requests	each tested, deployed, and verified against production
Events listed	10,240	official directory, Rock Star Librarian, community guides, camp-posted
Exodus crowd reports	9	the detector's entire first-season dataset – see limitations
Downtime	0	

THE TRAFFIC ENGINEER

The most consequential road in Black Rock City has no loop detectors, no cameras, and no published feed. We verified this exhaustively: the event’s public API offers camps, art, events and vehicles – no traffic endpoint; GARS 95.1 is analogue radio with no stream; the official traffic account on X sits behind a paid API and, it turned out, had been dormant since 2023. **For the corridor that produces six-to-nine-hour queues, there is no instrumentation at all.**

The crowd as detector

So the map treated its own users as probe vehicles. Anyone in the exodus line could tap one anonymous button – “Gate Road is taking about this long” – on a coarse scale from “moving” to “8 h or worse.” One report per person per half hour; the median of a rolling two-hour window became the corridor’s measured travel time. Coarse self-reports and NAT-shared phones make this a blunt instrument, but it is an *honest* one, and it is the only instrument the corridor has.

Level of Service, worn by the road

Measured travel time was banded against the corridor’s ~45-minute free-flow run (Greeters to pavement, ~8 miles of washboard at 10 mph pulses), A through F, spanning the documented 6–9 h historical peak:

GRADE	MEDIAN TRAVEL TIME	DISPLAY
A	≤ 45 min	green – free flow
B	≤ 1½ h	light green – moving well
C	≤ 2½ h	yellow – slow and steady
D	≤ 4½ h	amber – heavy
E	≤ 6½ h	orange – severe
F	beyond	red – gridlock

On the homepage, the entire surveyed corridor – the in-city stub and the run to the pavement – breathed slowly in its grade’s colour, and a procession of taillights (one per open rideshare post) crawled outward at a pace proportional to the reported travel time. A variable message sign, in effect, rendered as ambience.

The engineering rule we would defend anywhere: a grade computed from stale detectors is *withheld, not dimmed*. Two hours without a fresh report and the corridor’s colouring vanishes

entirely. Detectors down must never look like conditions green.

The 2026 corridor itself validated the band design: press and traveller reporting recorded **8¼-hour entry waits on August 30**, exodus peaks **over five hours** on the first big departure night, easing to 2-3½ hours in the small hours of September 7 and to ~30 minutes by mid-morning – the full A-through-F span, traversed in one weekend.

Honest result: the instrument shipped late – mid-exodus – and collected exactly **nine crowd reports** this year. The contribution of 2026 is a proven detector design and display grammar, not a season of data. Next year it deploys before gate opens; the recommended upgrade is opt-in GPS breadcrumbing from consenting phones in the line, which converts self-estimates into measured segment times with no new hardware.

THE PARTICIPANT

From a dusty phone, the map answered the questions that actually occur out there, in the order they occur:

- **"Where am I?"** – live GPS reverse-geocoded to the city's own address grammar ("*near 7:30 & Eternal*"), on a tile-free map that rendered entirely offline once loaded. No signal required, ever, for the map itself.
- **"Where are my people?"** – a Meshtastic LoRa radio paired over Bluetooth put campmates on the map and carried text chat with *no internet at all*; peers persisted across reloads as a "where I last saw them" view.
- **"What is the sky about to do?"** – a status pill led by sensors physically inside the fence (BRC NOC first), with the dust risk written in words – "*Dusty – goggles up*" – and the map itself animating the actual gusts and, when the model said so, rain. The week gave it material: 50 mph gusts tore through opening weekend and the first day neared 100 °F.
- **"What's on?"** – ~1,700 events; BMIR and Shouting Fire one tap away, on a player that kept playing across every page.
- **"How do I get home?"** – a rideshare board where offers stated seats, requests stated luggage, and a mutual-consent **Connect** shared live locations between exactly two people until either tapped stop – at which moment both positions were deleted, not hidden. The board also mirrored ridefinder.site's public listings, read-only and attributed, so one look showed the whole picture. The board's very first post was someone in a genuinely hard spot trying to reach Reno – and it drew a connection request within days. The feature existed for exactly that person.
- **"What do I owe the playa?"** – as Resto began, anyone standing on matter out of place could pin it anonymously, category-coded, for the line sweeps to find.

And one thing the participant never saw: an advertisement, a tracking pixel, or a consent banner – because there was nothing to consent to.

THE ORGANIZATION

BRC Map is unofficial and says so on every surface. From the Org's side of the fence, the season demonstrated three things:

Their public data, made legible

The placement team's directory stores camp locations as strings – "7:30 & E" – with no coordinates. Our parametric geocoder placed 1,140 of 1,184 of them (96%), refusing the rest by name (portals, Airport Road) rather than guessing; where a camp had also placed itself, the two positions agreed within 150 m in 99% of cases, median 37 m. Their art directory, which does carry GPS, went from paperwork to 351 pins in an afternoon. The standing rule – **a pin placed by a person always beats the spreadsheet** – meant the mirror never overwrote a single human placement.

Ten Principles, load-bearing

Decommodification and Leave No Trace are constraints we engineered against: no ads, no trackers, analytics that cannot identify anyone, a tip jar kept to one quiet footer line – and a Resto tool whose entire purpose is the MOOP Map's success. Radical self-reliance got an offline-first PWA; communal effort got a map drawn from the community's own pins.

A respectful request

The single highest-value thing the Org could publish is a **machine-readable Gate Road / Exodus feed** – even a manually updated JSON of "current travel time from Greeters." GARS reads it on air already; the same number, at a URL, would let every community tool (ours and others') show it to the people who cannot yet hear 95.1. Second on the list: coordinates in the camp placement API, which their own art API already proves is possible.

LAW ENFORCEMENT & PUBLIC SAFETY

Written plainly, because this audience reads for what a system *can* do, not what it promises.

What the system contributes to safety

- **Fewer people lost:** offline GPS with human-readable addresses means a disoriented participant at 3 a.m. can tell a Ranger where they are in the city's own vocabulary.
- **Fewer stranded travellers:** the rideshare board moves passengers into empty seats before they become roadside problems on SR-447; its first real user was a person leaving a bad situation, given a visible channel to the whole community rather than hitchhiking blind.
- **Exodus flow:** honest wait information reduces the surge-when-it-looks-low behaviour the Org itself documents; our display deliberately goes silent rather than show a stale low number that would trigger a rush.
- **Emergency locatability:** medical, Ranger and ESD stations are mapped landmarks; during the pre-event wildfires the Live page carried NWS alerts verbatim with explicit "we are not an emergency service" framing and links to the agencies that are.
- **Off-grid continuity:** Meshtastic positions and messaging function with all infrastructure down – a property that matters most in exactly the scenarios public safety plans for.

What the system holds – the part to read twice

There is, by architecture, almost nothing to request. Visit counting stores a salted hash that rotates at playa midnight; it cannot be reversed to an IP and cannot connect one day to the next. The rideshare Connect feature stores only the *current* position of two consenting parties and deletes both the moment either stops – there is no location history table in the schema at all. Anonymous exodus and MOOP reports carry the same rotating hash. Camp and art pins are voluntarily published by their owners and are already public. The honest summary for any records request: accounts (email, display name, their own posts) exist; movement histories do not, for anyone, ever – not as policy, but as the absence of the tables that would hold them.

We consider this a public-safety *feature*: a community that trusts its tools reports hazards, marks MOOP, and asks for rides through visible channels instead of improvising invisibly.

LIMITATIONS, STATED WITHOUT VARNISH

- The exodus detector and MOOP tool shipped mid-event; their 2026 datasets – nine wait reports, zero MOOP pins as of this revision – are proofs of machinery, not of adoption.
- Visitor counts are floors: every phone behind one camp's Starlink counts once.
- Meshtastic usage instrumentation began on day six and has observed a single radio – the panel works; adoption is an open question for next year.
- The rideshare board hosted one post and one connection in its first days – launched the night before exodus, it was infrastructure for a demand it hadn't yet met.
- The official traffic account mirror works only when X permits, which is rarely; it degrades to a link by design.
- One partner station (K-Hole 102.3) was never once observed streaming; its player was removed rather than left apologising.
- Nothing in this report was verified visually by its primary author, which is the kind of sentence that belongs in a limitations section.

RECOMMENDATIONS FOR 2027

- Deploy the exodus detector before gate opens; add opt-in GPS breadcrumbs for measured segment times.
- Pursue the RideFinder partnership (contact begun) toward a shared feed, so one post reaches both boards.
- Restore a lightweight artist-claim flow before the next art import; this year's import left 100+ official artworks permanently ownerless.
- Generalise the crowd meter: the report/median/freshness pattern fits ice lines, fuel, and medical waits unchanged.
- Ask the Org, formally, for the Gate Road feed above.

BRC Map (brcmap.net) · GPL-3.0-or-later · built by burners in partnership with an AI pair (Claude), whose full account appears in the project README · Pins are approximate; only Burning Man Placement determines camp locations and only the ARTery determines art placement · This report contains no personal data.